

Automating a first reproducible bibliometric analysis to support HCERES-style evaluation preparation.

Problem

Preparing an HCERES-style evaluation requires a reliable synthesis of scientific output, collaborations, publication venues, and open science indicators.

- Most of the data already exists in HAL, but metadata is heterogeneous.
- Manual extraction is slow, hard to reproduce, and difficult to audit.
- The objective is not a final institutional report, but a verifiable working basis.

Objective

Build a local, deterministic, and rerunnable pipeline to produce a first bibliometric analysis of the ACENTAURI team from HAL.

Inputs

- HAL query or structure identifier;
- analysis period;
- optional CSV files: members, themes, Scimago, CORE.

Outputs

- CSV and LaTeX tables;
- figures;
- Markdown report;
- standalone LaTeX report.

Prompt reconstructed from the repository

Create a simple, monolithic Python proof of concept to automate a HAL-based bibliometric analysis of the ACENTAURI team in a format useful for HCERES preparation.

- Query HAL automatically, with structure discovery or explicit override.
- Clean, filter, and deduplicate publications.
- Compute indicators: output, authors, document types, collaborations, venues, open science.
- Accept optional CSV files as enrichments, without making them blocking dependencies.
- Generate deterministic reports, with no paid LLM API and no invented content.

Note: the exact prompt is not versioned in the repository; this slide summarizes the implicit specification reflected by the generated code.

- ① Discover the HAL query or use an explicit structure identifier.
- ② Retrieve HAL records through paginated API calls.
- ③ Normalize metadata: title, authors, dates, types, affiliations, DOI, files.
- ④ Filter out non-target records: theses, HDR, reports, lectures, undefined categories.
- ⑤ Deduplicate records and identify ACENTAURI members.
- ⑥ Compute indicators and export deliverables.

Design principle

The script distinguishes verified HAL metadata, inferred information, manually supplied data, and unavailable information.

Existing run results

- 69 HAL records retrieved;
- 58 publications retained;
- 31 members identified or inferred;
- Markdown, LaTeX, and PDF reports generated.

To consolidate

- validate the member list;
- provide the ARC/RIC/MOC mapping;
- add Scimago and CORE data;
- review partner classifications.

Next step

Move from POC to reliable deliverable by versioning validation CSV files and adding domain-level quality control before institutional use.